PHASE 0

Foundations

A simple, step-by-step booklet

What Phase 0 builds, what it leaves alone, and what you will see.

0 1 2 3 4 5 6 7 8

9 small steps

2026-11-09

Mock-ups marked 'expected' show the intended result.

What is inside

3	The big picture
4	The road ahead
5	What Phase 0 builds and leaves alone
6-14	Steps 0 to 8, one per page
15	The flow in one line
16	What Phase 0 does NOT do
17	Files it touches
18	Check it yourself in 5 minutes
19	Phase 0 is done when

Phase 0 in one sentence

Phase 0 builds the **plumbing** that every later phase needs.

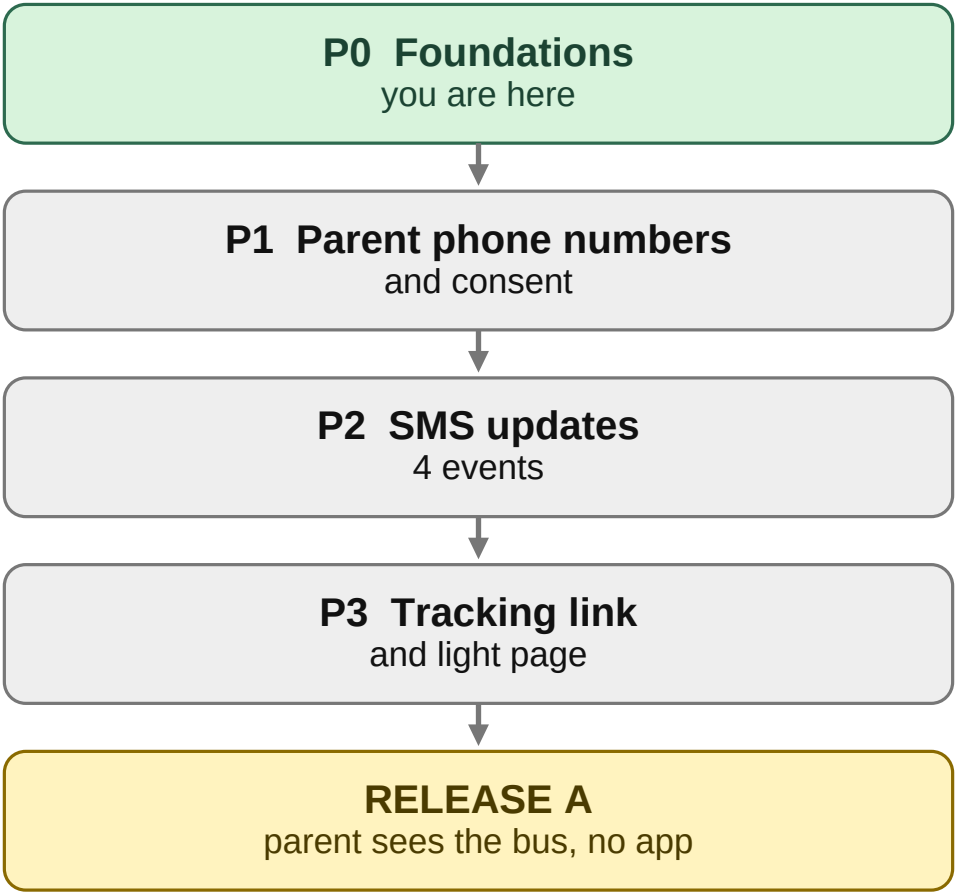
A per-school on/off switch, a phone-number cleaner, safe logging, Nairobi time, a place to run background jobs, and a health check. It also fixes the admin dashboard's wrong counts.

THE IMPORTANT BIT

No parent will see anything yet. No school console screen changes.

The only visible change is that the admin dashboard counts become correct.

The road ahead



Later: P4 WhatsApp, P5 M-Pesa, P6 Admin, P7 Pilot, P8 Device and verified boarding.

Builds and leaves alone

PHASE 0 BUILDS

- Per-school feature switches
- Phone-number cleaner
- Safe logging
- Nairobi time helper
- Redis queue for new jobs
- Health check command
- Config key names
- Dashboard count fix

PHASE 0 LEAVES ALONE

- Driver app and Guardian app
- Push notifications
- Every existing endpoint
- users/all-schools (only written up)
- Server time zone (stays UTC)
- The .env file (never read)

0 Baseline commit and test run

WHAT IT DOES

Saves a first snapshot of the project (without the .env secrets file) so every later change can be reviewed. Then runs the existing tests to record the starting score.

YOU WILL SEE

A before-and-after test result: pass and fail counts.

HOW IT FLOWS

1 Dashboard count fix

WHAT IT DOES

The admin dashboard counts drivers, parents and students using the wrong role numbers. One file, AdminDashboardController, is corrected.

YOU WILL SEE

Correct counts on the admin dashboard.

Label	Today	After P0
Schools	2 (right)	2
Drivers	**2 (wrong)**	**3**
Parents	**3 (wrong)**	**4**
Guardians	5 (right)	5
Students	**4 (wrong)**	**6**

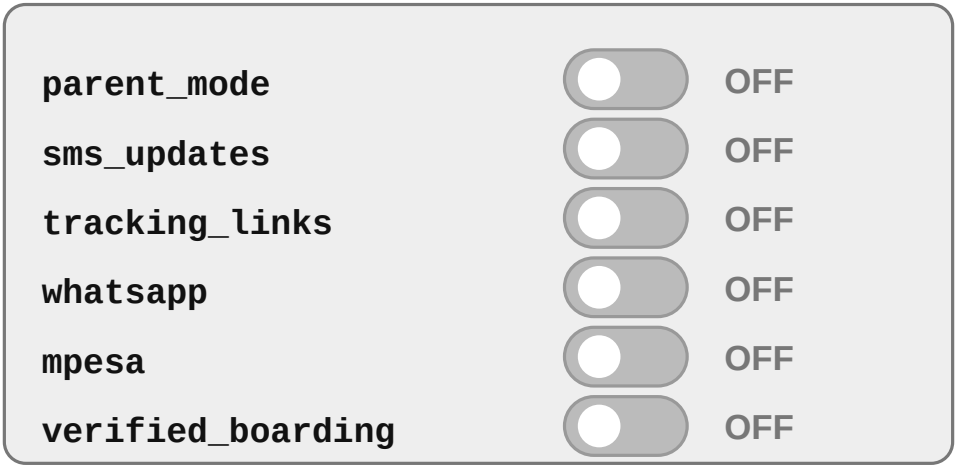
2 school_features: the switches

WHAT IT DOES

A table of per-school on/off switches. All OFF by default. Only admins can flip them, and every flip is recorded.

YOU WILL SEE

An admin-only API to read and flip switches. No screen yet.



3 PhoneNumber helper

WHAT IT DOES

Turns any Kenyan number format into one standard +254 form, rejects bad numbers, and hides the middle digits when shown. Existing numbers are not rewritten.

YOU WILL SEE

Examples below, backed by 30+ tests.

You type	Kept	Shown
`0712345678`	+254712345678	+254712***678
`712345678`	+254712345678	+254712***678
`254712345678`	+254712345678	+254712***678
`071234567`	REJECTED	too short
`0612345678`	REJECTED	not mobile

4 Safe logging

WHAT IT DOES

New outgoing web calls log their activity with phone numbers masked and secrets removed. Old code that logs device tokens is not edited, only listed for you.

YOU WILL SEE

Log lines like the SAFE example below.

UNSAFE: must never appear

```
to=+254712345678 key=sk_live_AbC123
```

SAFE: what Phase 0 produces

```
to=+254712***678 key=[hidden]
```

5 SchoolTime: Nairobi clock

WHAT IT DOES

One place that gives Nairobi time to new code. The server itself stays on UTC, so nothing existing shifts.

YOU WILL SEE

Correct school-day boundaries in later phases.

6 Redis queue for new jobs

WHAT IT DOES

New background jobs (SMS, tracking links) get a real queue. Existing code keeps today's behaviour and runs immediately.

YOU WILL SEE

A supervisor config example in the docs folder.

OLD

Existing code runs immediately. Unchanged.

NEW

New jobs go: Redis queue, then worker.

7 Config key names

WHAT IT DOES

Names for the WhatsApp, SMS, Daraja (M-Pesa) and short-link settings, with no values filled in.

YOU WILL SEE

New lines in .env.example.

WhatsApp
names only

SMS
names only

Daraja (M-Pesa)
names only

Short links
names only

8 Health check command

WHAT IT DOES

One command that says whether Redis, the queue worker and the scheduler are alive, and names the broken one if not.

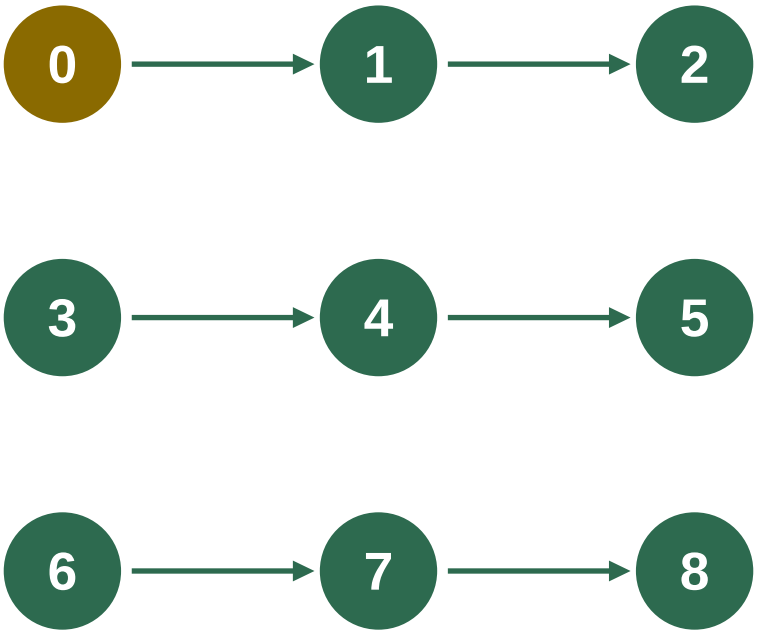
YOU WILL SEE

Output like below. Command name is a proposal.

```
$ php artisan health:check
Redis ...... OK
Queue ....... OK
Scheduler ... OK
All checks passed.
```

```
$ php artisan health:check
Redis ...... OK
Queue ....... FAIL
Scheduler ... OK
1 problem found.
```

The flow in one line



Step 0 comes first so every later change is measured against a known starting point.

Steps 1 to 8 are small, separate and additive, so a problem in one cannot break the others.

What Phase 0 does NOT do

- No change to the driver app, guardian app, push notifications or any existing endpoint.
- users/all-schools is not touched. Phase 0 only writes up what it exposes.
- No messages sent, no parent screens, no tracking page, no M-Pesa.
- Server time zone stays UTC. The .env file is never read.

Files it touches

Expected. Confirmed in the plan step before anything is built.

NEW

- school_features migration and model
- PhoneNumber and SchoolTime helpers
- Health check command
- Log-masking step for web calls
- Supervisor config example (docs)
- Tests for each of the above

EDITED (small, additive)

- AdminDashboardController.php (count fix)
- routes/api.php (one admin-only group)
- Console/Kernel.php (one heartbeat)
- AppServiceProvider.php (web-call step)
- .env.example (key names only)

Check it yourself in 5 minutes

1. Open the admin dashboard. Driver, parent and student counts match real users.
2. Ask for one school's switch list. All six read OFF.
3. Run the health command. All OK, or it names the broken one.
4. Look at the test report. All 30+ phone examples pass.
5. Compare test results before and after. Same or better.

Phase 0 is done when

- [ ] Baseline snapshot exists, .env not included
- [ ] Test results recorded before and after
- [ ] Dashboard counts correct, covered by tests
- [ ] Six switches: default off, admin-only, audited
- [ ] 30+ phone tests pass
- [ ] Logs show masked numbers only
- [ ] Health command works and reports a failure
- [ ] No existing endpoint or app behaviour changed
- [ ] Report under 500 words delivered to you

Notes